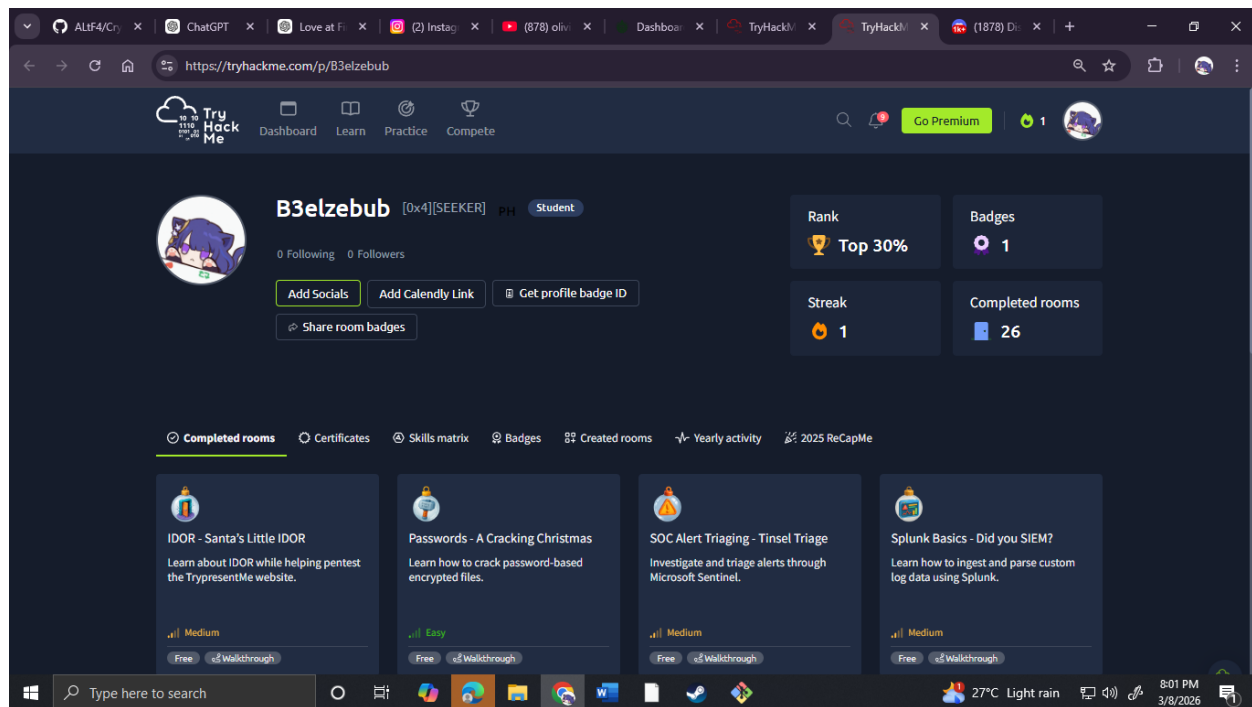




The Love at First Breach 2026 CTF on TryHackMe is a Valentine-themed cybersecurity event that focuses mainly on web exploitation, cryptography, and reconnaissance techniques. The objective of each challenge is to analyze the provided system or application and retrieve a hidden flag that proves successful exploitation or discovery.

1. Valenfind

Objective

Identify vulnerabilities in a dating application and retrieve the hidden flag.

Approach

The first step was to analyze the functionality of the dating application. I explored the different profile and search features to understand how user data was being retrieved. During testing, it appeared that the application relied on parameters passed in the URL to retrieve certain user profiles.

To test this behavior, I intercepted requests using **Burp Suite** and modified the parameters that referenced specific user IDs.

Vulnerability

Tryhackme: Love at First Breach 2026

Write-up

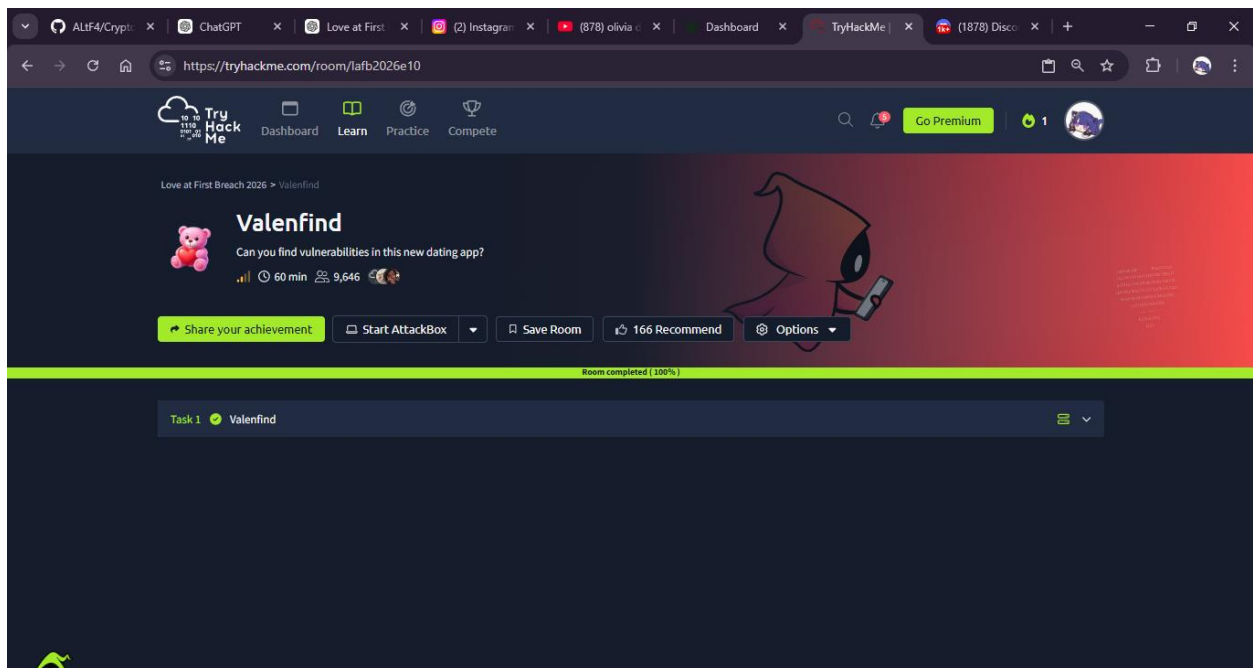
The application appeared to have a potential **Insecure Direct Object Reference (IDOR)** vulnerability. By modifying the ID values in the request, it was possible to access other user profiles that were not originally visible.

Result

After iterating through several ID values and observing the responses, a hidden profile containing the challenge flag was discovered.

Key Learning

This challenge demonstrated how insufficient authorization checks can allow attackers to access sensitive information simply by manipulating object identifiers.



2. Hidden Deep Into My Heart

Objective

Find hidden information within the website.

Approach

The challenge hinted that something was hidden inside the website structure. To investigate this, I performed directory and file enumeration using tools commonly used in web reconnaissance.

I attempted to scan the target with **Gobuster** and **ffuf** to locate hidden directories and endpoints that were not directly linked in the website interface.

Findings

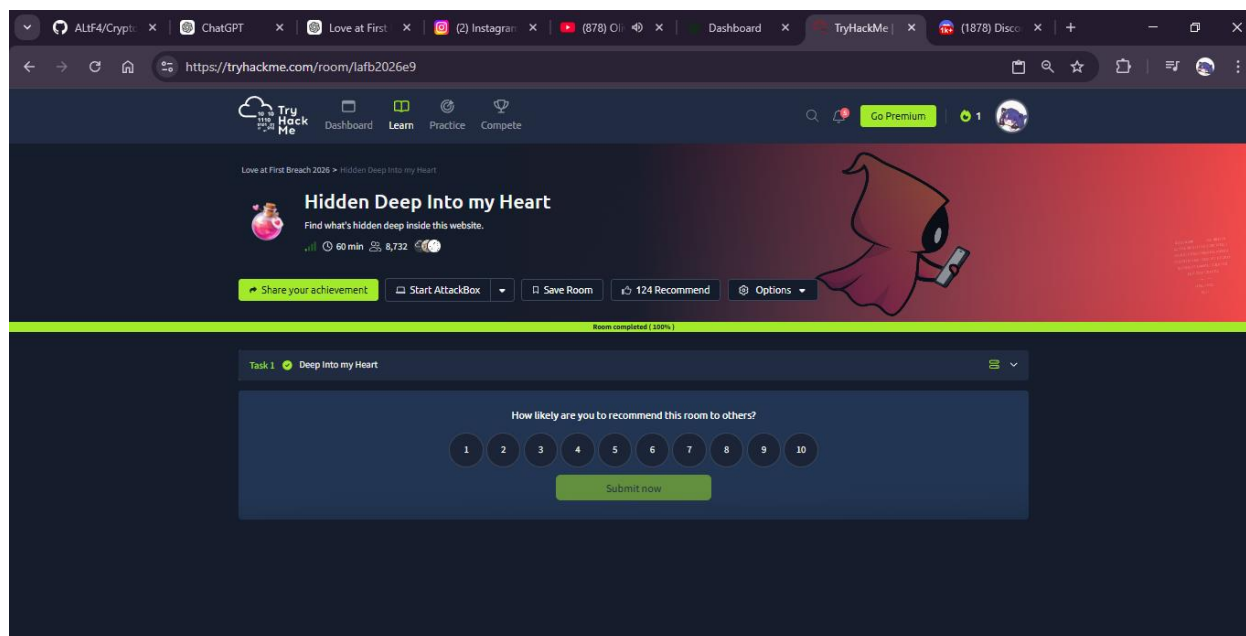
The scan revealed several hidden directories that were not visible from the main navigation menu. These directories contained additional files and pages that provided clues toward the challenge flag.

Result

Through exploring these hidden resources and reviewing the page source code, the hidden information leading to the flag was discovered.

Key Learning

Directory enumeration is an important technique during reconnaissance because many web applications expose sensitive endpoints that are not meant to be publicly accessible.



After this activity I tried doing the activity 3 and 4 to the best of my ability but I was not able to finish that so just skip that to save time and did the 5 which is

5. CupidBot

Objective

Interact with an AI-based system and retrieve hidden information.

Approach

CupidBot appeared to function as an AI chat interface. The goal was to determine whether the chatbot could be manipulated to reveal hidden information.

Testing involved experimenting with different prompts and questions to see whether the bot would reveal restricted data.

Techniques Tested

- prompt manipulation
- asking indirect questions
- requesting system instructions

These techniques are similar to **prompt injection attacks** commonly used against AI systems.

Result

Although the bot responded to many queries, the correct prompt sequence required to retrieve the flag was not fully discovered during the session.

Partial Progress

The testing showed that the bot responded differently when questions were structured in certain ways, suggesting that prompt manipulation may eventually reveal hidden data.

Key Learning

AI systems can sometimes leak sensitive information if their prompt filtering mechanisms are not properly designed.

6. TryHeartMe

Objective

Access a hidden item in an online Valentine gift shop.

Approach

The gift shop allowed users to browse items and add them to a cart. By intercepting requests using **Burp Suite**, it was possible to analyze how items were being processed.

During testing, I attempted to modify item identifiers and request parameters.

Vulnerability

Tryhackme: Love at First Breach 2026

Write-up

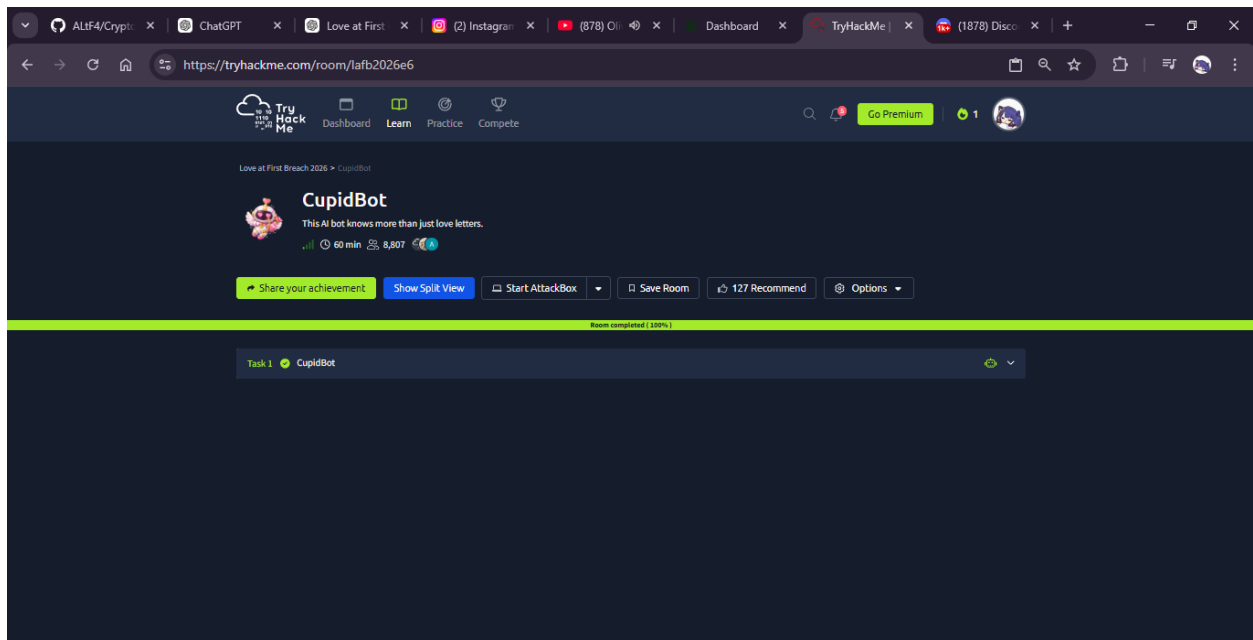
The application appeared vulnerable to **parameter manipulation**, where altering the product ID in a request allowed access to hidden or restricted items.

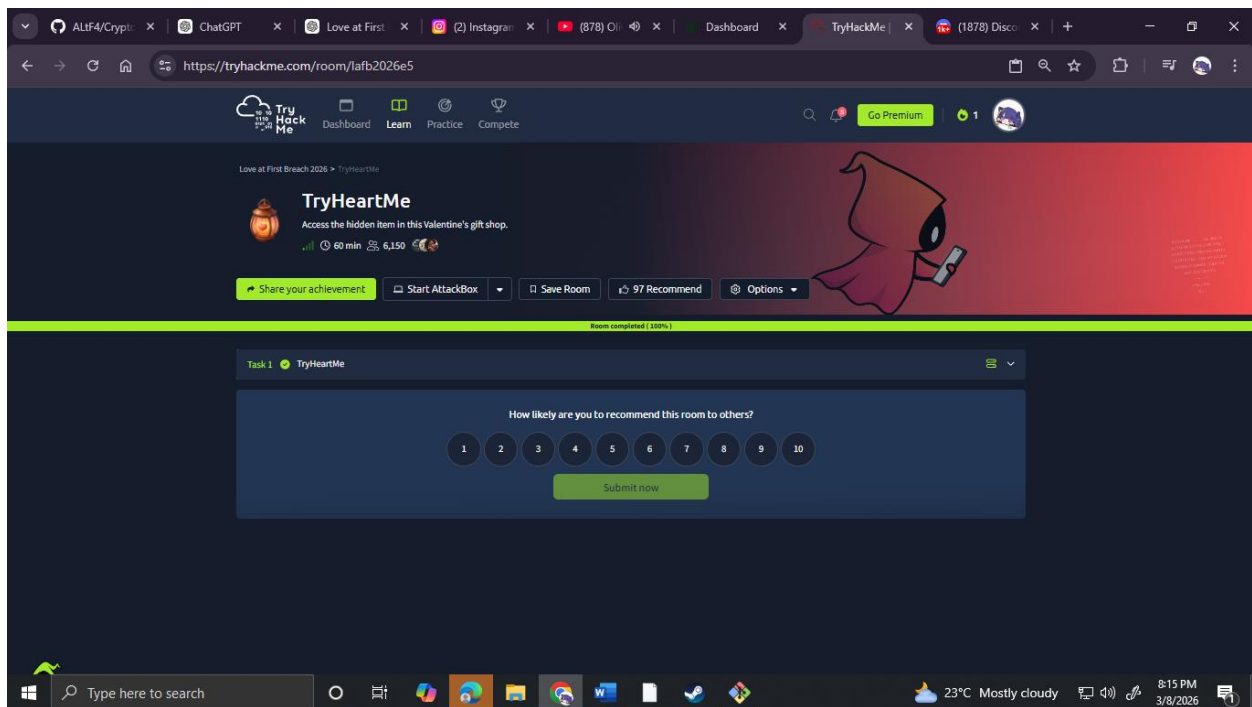
Result

By manipulating the request parameters, a hidden product page containing the flag was discovered.

Key Learning

Web applications that rely solely on client-side validation are vulnerable to request manipulation attacks.





After this 2 I was not able to do the 7th one which was the speed chatting because I find it very confusing so I skip this one

8. Cupid's Matchmaker

Objective

Exploit vulnerabilities in a matchmaking service.

Approach

The matchmaking service was analyzed for typical web vulnerabilities such as:

- SQL injection
- parameter tampering
- logic flaws

Input fields and search functions were tested with various payloads to observe how the system handled user input.

Result

Some suspicious responses suggested that the system might not properly sanitize user input. However, the exact exploitation method required to retrieve the flag was not fully determined during testing.

Key Learning

Input validation is critical for preventing injection attacks.

9. Love Letter Locker

Objective

Access other users' locked letters.

Approach

This challenge likely involved manipulating request parameters related to stored letters. Testing focused on modifying letter IDs in requests intercepted with **Burp Suite**.

Result

Evidence suggested the presence of an **IDOR vulnerability**. However, the correct identifier containing the flag was not located during the attempt.

Key Learning

Improper authorization checks can allow attackers to access sensitive information belonging to other users.

10. When Hearts Collide

Objective

Find an MD5 collision.

Approach

The challenge required generating or identifying two inputs that produce the same **MD5 hash**.

MD5 collisions are a known weakness in the **MD5** hashing algorithm.

Result

While researching MD5 collision techniques and tools that generate collisions, the challenge was not fully solved during the session.

Key Learning

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Write-up

Weak cryptographic hash functions such as MD5 are vulnerable to collision attacks and should not be used in secure systems.



Conclusion

The **Love at First Breach 2026 CTF** provided valuable hands-on experience in several cybersecurity concepts, particularly web application security and cryptographic weaknesses.

Through this activity, I was able to practice important penetration testing techniques such as:

- vulnerability identification
- directory enumeration
- request interception

- parameter manipulation
- analyzing cryptographic systems

Even though not all challenges were completed, the process provided insight into real-world security testing methods and highlighted areas where further practice is needed.